



METAL SEGMENTATION FOR ARTIFACT REDUCTION

Projection-Domain vs Image-Domain Segmentation

A comparative study of two deep-learning approaches to segmenting metal implants in CT / CBCT scans, with a hands-on U-Net implementation.

Agenda

What we'll cover today

The Problem



Why metal implants corrupt CT/CBCT images and why segmentation matters

Two Domains



Image-domain vs projection-domain metal segmentation

Paper 1 — CBCT Projection Domain



Synthetic-data U-Net for projection-domain segmentation (Agrawal et al.)

Paper 2 — CT Image Domain



Diffusion-based segmentation, DiffSeg (Xie et al.)

Head-to-Head Comparison



Which domain segments metal better, and why

My Implementation



U-Net based metal mask segmentation notebook — approach & results

The Problem: Metal Artifacts in CT / CBCT

Why accurate metal segmentation matters

Metal implants (screws, plates, dental fillings, hip prostheses) are high-density and cause severe artifacts:

- Blooming, streaking and shading around metal regions
- Obscured boundaries hinder diagnosis & dose planning
- Metal Artifact Reduction (MAR) inpaints the metal trace — but this requires accurate metal segmentation first
- Manual segmentation is slow, subjective, and impractical for thousands of CT/CBCT projections

Typical MAR Pipeline

1. Acquire CT / CBCT scan
2. Segment metal
(image domain or projection domain)
3. Mask / inpaint metal trace
in projections or sinogram
4. Reconstruct corrected image

Step 2 — metal segmentation — is the focus of both papers reviewed.

Two Segmentation Domains

Where in the imaging pipeline does segmentation happen?



Image Domain

Segmentation is performed on the reconstructed 3D CT volume (axial slices), typically using HU-based thresholds or learned models on reconstructed images.

- Operates on reconstructed slices (e.g. 512x512 HU volumes)
- Simple thresholds (2500/3000 HU) are common baselines
- Must be forward-projected back to sinogram/projection space for projection-based MAR



Projection Domain

Segmentation is performed directly on the raw 2D X-ray projections (sinogram for CT, projections for CBCT) before reconstruction.

- Operates directly on raw 2D projections / sinograms
- No forward-projection step needed for MAR
- Handles motion blur and out-of-FOV metal naturally

Paper 1 — Projection-Domain Metal Segmentation (CBCT)

Agrawal, Hietanen & Sarkka, IEEE Access 2023



Synthetic Training via Monte Carlo Simulation

- Real ground-truth metal labels are hard to obtain — boundaries are unclear and there are hundreds of projections per scan
- Solution: simulate metal-corrupted CBCT projections + ground truth from 10 real metal-affected clinical scans (knee, wrist, foot, ankle, palm, forearm)
- Compares noisy (300-1400 photons/pixel) vs clean (5000 photons/pixel) simulations — noisy is cheaper & more realistic
- Trains a modified U-Net (4 down/up blocks, instance norm) on full-size projections + crops together for robustness

Key Results (10 clinical test scans)

94.8%

Mean DSC
Best model: Noisy (full+crops)

90.2%

Mean IOU
Noisy (full+crops)

90.5%

Mean DSC — CMAR
(threshold-based baseline)

0.51e-3

False positive rate
on 6 metal-free scans

Best result: Noisy simulation + full-size & cropped projections trained together significantly outperformed CMAR, MMAR, and consistency-check methods, especially for motion-affected and out-of-FOV metals.

Paper 1 — Model & Training Pipeline

Modified U-Net trained purely on simulated CBCT projections



Network: U-Net variant — 16 to 256 channels, 3x3 conv + ReLU + instance norm, bilinear upsampling, skip connections, 1x1 output conv + sigmoid. Trained on a single RTX 2080 Ti, Adam optimizer, BCE loss, with stacked augmentations (flips, elastic deformation, noise, dropout).

Paper 2 — Image-Domain Metal Segmentation (CT)

Xie et al., DiffSeg — BMC Medical Imaging 2024



Diffusion Probabilistic Model for Metal Masks

- Operates on reconstructed axial CT slices (512x512), not raw projections
- DiffSeg = ResU-Net diffusion model + Conditional Dynamic Coding (fuses mask + prior image) + GFParser (removes high-frequency noise via learnable Fourier filter)
- Trained on 100 patients (11,280 slices) with simulated metal artifacts inserted into real artifact-free CT images
- Validated on clinical artifact CT, CTPelvic1K, and physical phantoms with known titanium rod geometry

Key Results (2,820 test slices)

95.45%

Mean DSC — DiffSeg
(best among all models)

97.89%

Accuracy — DiffSeg

84.19%

Mean DSC — T3000
threshold baseline

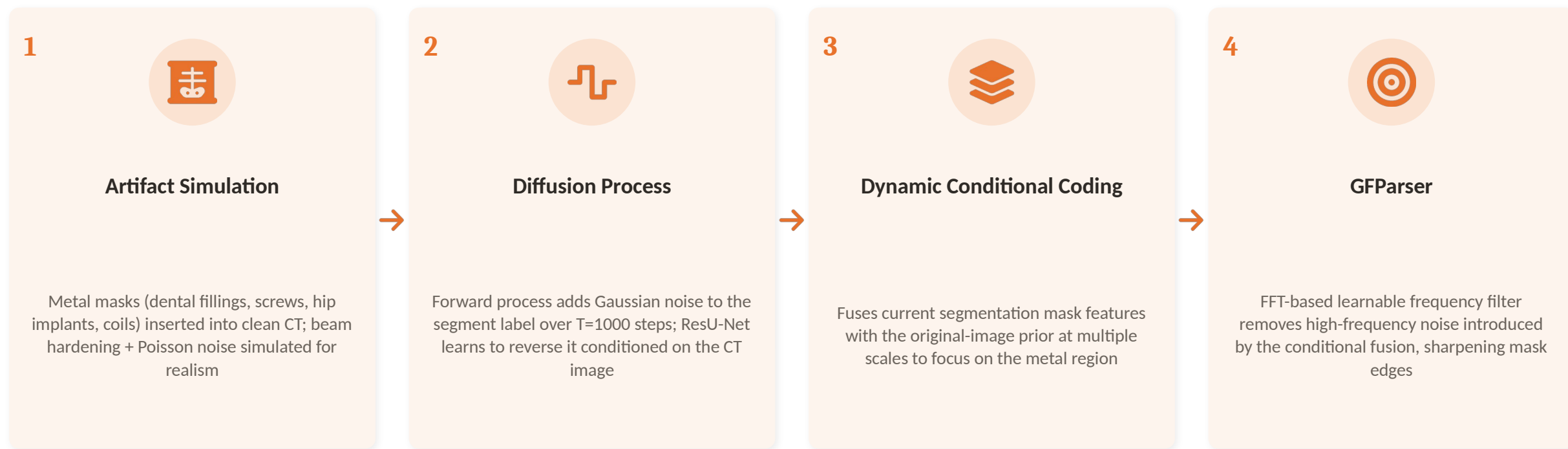
93.52%

Mean DSC — U-Net
(traditional baseline)

DiffSeg outperformed U-Net, Attention U-Net, R2U-Net & DeepLabV3+ on simulated, clinical, and phantom data (statistically significant, $p < 0.05$ vs most models). Its smaller, tighter masks better preserve surrounding bone detail during MAR.

Paper 2 — DiffSeg Architecture

Conditional dynamic coding + global frequency parser on a ResU-Net backbone



Ablation impact: Removing both modules drops DSC to 93.56% (DiffSeg_1). Dynamic conditional coding alone gives 94.35% (+0.79%). Adding GParser reaches the full 95.45% (+1.1% total) — each component contributes measurably.

Head-to-Head Comparison

Projection-domain (CBCT) vs Image-domain (CT) metal segmentation

Aspect	Projection Domain (Paper 1)	Image Domain (Paper 2)
Modality / Data	CBCT, extremity scans (knee, wrist, ankle, foot)	CT, head/neck, chest, abdomen + phantoms
Model	Modified U-Net (CNN, supervised)	DiffSeg — diffusion model + ResU-Net
Training data	Purely synthetic (Monte Carlo simulation)	Simulated artifacts on 100 real patient CTs
Best DSC	94.8%	95.45%
Best Accuracy / IOU	IOU = 90.2%	ACC = 97.89%
vs Threshold baseline	CMAR (region-growing): DSC 90.5%	T3000 threshold: DSC 84.19%
Key strength	No forward projection needed; robust to motion, out-of-FOV metal, and voxel-size changes	Sharper, smaller masks; better detail near metal in NMAR correction
Inference cost	Single forward pass (fast, ~ms)	Diffusion sampling (~3 s/batch, DPM-Solver)



VERDICT

Which Domain Segments Metal Better?

It depends on the goal — but **projection-domain segmentation is the stronger choice for CBCT-based MAR pipelines**, while image-domain (DiffSeg) wins on raw segmentation accuracy for reconstructed CT.



Projection Domain wins when...

- ✓ Metal is out-of-FOV or scan has heavy motion artifact (image-domain cannot segment what isn't reconstructed)
- ✓ Pipeline needs direct integration into existing projection-based MAR (no forward-projection step)
- ✓ Real-time / single-pass inference is required



Image Domain (DiffSeg) wins when...

- ✓ Highest raw DSC/Accuracy on reconstructed slices is the priority (95.45% vs 94.8%)
- ✓ Fine boundary detail and noise suppression matter (GFParse removes high-frequency artifacts)
- ✓ Sufficient compute is available for diffusion sampling (~3s per batch)

My Implementation — U-Net Metal Mask Segmentation

Task: build & train a U-Net to segment metal regions in CT slices



Approach

- Dataset: 500 CT slices (512x512) with paired binary metal masks, loaded as raw float32 arrays
- Preprocessing: min-max normalization, resized to 128x128 (image: bilinear, mask: nearest-neighbour to keep it binary)
- Model: lightweight U-Net — 2 encoder blocks (16->32 ch), bottleneck (64 ch), 2 decoder blocks with skip concatenation, 1x1 conv + sigmoid output
- Training: 80/20 train-test split, Adam optimizer, binary cross-entropy loss, 15 epochs
- Image-domain task — directly comparable to DiffSeg approach in Paper 2

Training Results

99.86%

Train accuracy
(epoch 15)

99.89%

Val accuracy
(epoch 15)

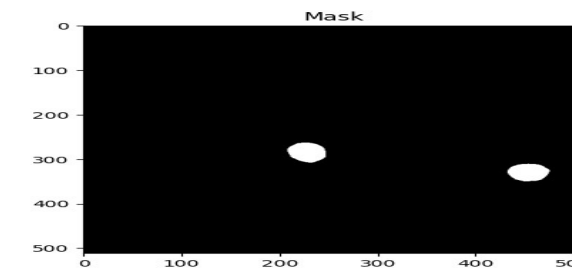
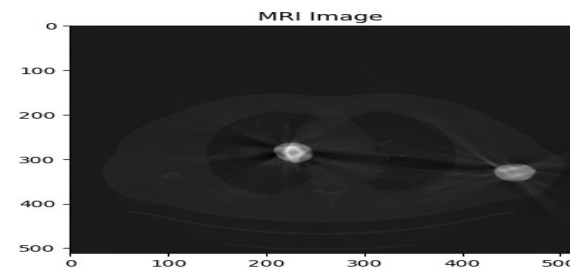
0.0040

Train loss (BCE)
(epoch 15)

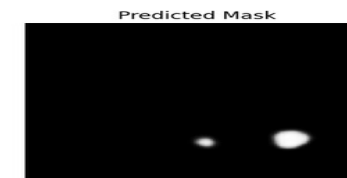
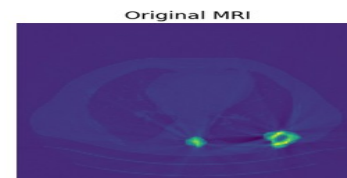
0.0033

Val loss (BCE)
(epoch 15)

Sample input slice & ground-truth metal mask



Predicted mask vs ground truth (test sample)



Implementation — Observations & Next Steps

Where this fits relative to the reviewed papers



Observations

- Validation accuracy (99.89%) is very high but driven mainly by background pixels — DSC/IOU on the metal class would give a fairer picture (as both papers report)
- Predicted mask is smoother / slightly larger than ground truth — similar to over-segmentation seen with simple threshold methods in Paper 2
- 128x128 downsampling may blur small or thin metal structures; both papers avoid downsampling near metal boundaries for this reason
- A compact 2-level U-Net was sufficient for this dataset's large, well-separated metal regions



Suggested Next Steps

- Report DSC / IOU / sensitivity / specificity per the metrics used in both papers for direct comparability
- Compare against a fixed-threshold baseline (2500/3000 HU) as done in Paper 2
- Apply data augmentation (flips, elastic deformation) following U-Net (Ronneberger et al.) and Paper 1's strategy
- Evaluate on full-resolution slices / crops to preserve fine metal boundaries, as recommended in Paper 1
- Long term: extend to the projection domain (CBCT) using simulated data, mirroring Paper 1's pipeline



Summary

- ✓ Both domains push U-Net beyond simple HU thresholds — DSC improved from ~84% (threshold) to ~95% (deep learning)
- ✓ Projection-domain (Paper 1) is best suited to CBCT pipelines: motion-robust, handles out-of-FOV metal, single fast pass
- ✓ Image-domain DiffSeg (Paper 2) achieves the highest raw segmentation accuracy via diffusion + frequency-domain refinement
- ✓ My U-Net notebook implements the image-domain approach — a strong foundation to extend toward Paper 2's evaluation rigor

Thank you — questions welcome